Lab 4.2: Error Handling and Unit Testing

lab4.2.go

```go
package main

import (
    "fmt"
    "math"
    "strconv"
    "strings"
)

func Divide(a, b float64) (float64, error) {
    if b == 0 {
        return 0, fmt.Errorf("cannot divide %.2f by zero", a)
    }
    return a / b, nil
}

func Sqrt(x float64) (float64, error) {
    if x < 0 {
        return 0, fmt.Errorf("cannot take square root of negative number %.2f", x)
    }
    return math.Sqrt(x), nil
}

func ParseAge(input string) (int, error) {
    input = strings.TrimSpace(input)
    if input == "" {
        return 0, fmt.Errorf("input is empty: please enter an age")
    }
    age, err := strconv.Atoi(input)
    if err != nil {
        return 0, fmt.Errorf("invalid age %q: not a whole number", input)
    }
    return age, nil
}

func main() {
    if result, err := Divide(10, 2); err != nil {
        fmt.Println("Error:", err)
    } else {
        fmt.Println("Divide(10, 2) =", result)
    }
    if _, err := Divide(10, 0); err != nil {
        fmt.Println("Error:", err)
    }

    if result, err := Sqrt(16); err != nil {
        fmt.Println("Error:", err)
    } else {
        fmt.Println("Sqrt(16) =", result)
    }
    if _, err := Sqrt(-9); err != nil {
        fmt.Println("Error:", err)
    }

    if age, err := ParseAge("21"); err != nil {
        fmt.Println("Error:", err)
    } else {
        fmt.Println("ParseAge(\"21\") =", age)
```

lab4.2.go (contd.)

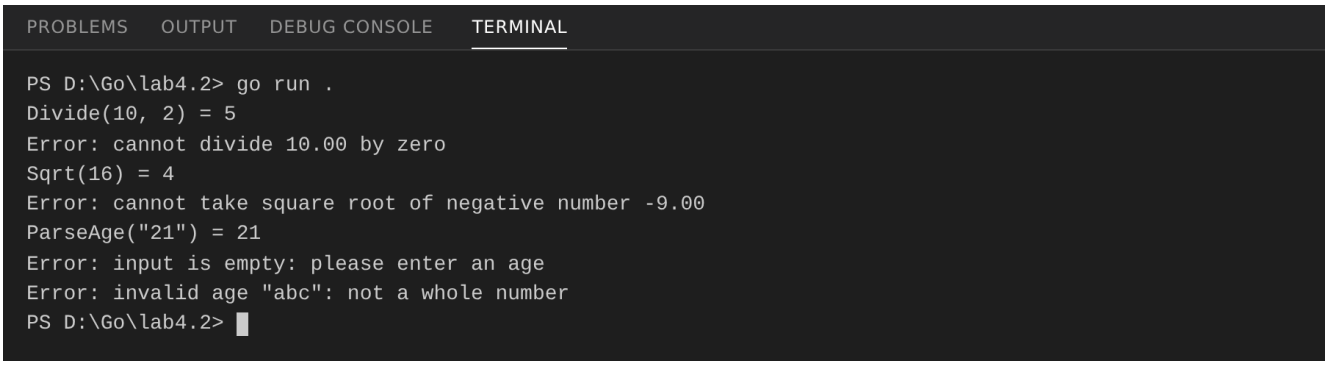
```
59      }
60      if _, err := ParseAge(""); err != nil {
61          fmt.Println("Error:", err)
62      }
63      if _, err := ParseAge("abc"); err != nil {
64          fmt.Println("Error:", err)
65      }
66  }
```

lab4.2_test.go

```
 1  package main
 2
 3  import "testing"
 4
 5  func TestDivide(t *testing.T) {
 6      got, err := Divide(10, 2)
 7      if err != nil || got != 5 {
 8          t.Errorf("Divide(10, 2) = %v, %v; want 5, nil", got, err)
 9      }
10
11      _, err = Divide(10, 0)
12      if err == nil {
13          t.Error("Divide(10, 0) should return an error")
14      }
15  }
16
17  func TestSqrt(t *testing.T) {
18      got, err := Sqrt(16)
19      if err != nil || got != 4 {
20          t.Errorf("Sqrt(16) = %v, %v; want 4, nil", got, err)
21      }
22
23      _, err = Sqrt(-9)
24      if err == nil {
25          t.Error("Sqrt(-9) should return an error")
26      }
27  }
28
29  func TestParseAge(t *testing.T) {
30      got, err := ParseAge("21")
31      if err != nil || got != 21 {
32          t.Errorf("ParseAge(\"21\") = %v, %v; want 21, nil", got, err)
33      }
34
35      if _, err = ParseAge(""); err == nil {
36          t.Error("ParseAge(\"\") should return an error for empty input")
37      }
38
39      if _, err = ParseAge("abc"); err == nil {
40          t.Error("ParseAge(\"abc\") should return an error for non-numeric input")
41      }
42  }
```

Lab 4.2: Output

Program output (go run .)



Test output (go test -v)

